Rémi Imbach

Curriculum Vitæ

Civil status: Rémi Imbach, born December 7,1985 in Strasbourg
French citizen, married, one child.

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Quick overview

Last positions

2022 - now **Research Engineer**, Université de Strasbourg (UdS), then Inria[1], France

2020 - 2022 **Consultant**, independent; contracts with Waterloo Maple Incorporation, Canada, and European Science Foundation, France

2018 - 2020 **Assistant Professor**, Courant Institute of Mathematical Sciences (CIMS), New York University (NYU), New York, USA

Education

2013 **PhD** in **Computer Science**, UdS, France

Recent software

2025 **SCimBa**[2]: Python library for Physics Informed Machine Learning. Backends Pytorch or Jax.

2023 **PWPoly**[3]: C library for fast polynomial evaluation and root-finding, Python wrapper and Maple module

Keywords

Scientific: Physics Informed Machine Learning, numeric and symbolic computation, certified computation, polynomial root finding, computer arithmetic

Technologies: Jax, Pytorch, Python, C, C++, CMake, Maple, git, LaTeX, CI, Linux

[1] National Institute for Research in Digital Science and Technology
[2] https://www.scimba.org
[3] https://gitlab.inria.fr/gamble/pwpoly

Positions

since dec. 2024	**Research Engineer**, Inria Nancy - Grand Est, France, MACARON[4] group *Physics informed machine learning library* Collaborators: E. Franck and V.Michel-Dansac (Inria), M. Boileau (CNRS) Software project: `SCimBa`
2024 10 mois	**Research Engineer**, IRMA[5], UdS, France, MoCo[6] group *Radiative transfer: inverse problem for water turbidity measurment*, Collaborator: J. Aghili (UdS) Software project: `AnaTrot`(not published) Teaching: 35 hours, UdS
2022 - 2023 18 months	**Research Engineer**, Inria Nancy - Grand Est, France, GAMBLE[7] group *Implementation of a root-finding algorithm, integration in Maple*, Collaborator: G. Moroz (Inria) Software: `PWPoly` Publication: [1]
2020 - 2022 20 months	**Consultant**, independent; contracts with V. Pan (CUNY[8]), with WMI[9] and with ESF[10]. Collaborators: V. Pan, G. Moroz and F. Rouillier (Inria), J. Gerhard (WMI) Publications: [2, 3] Software: `Ccluster`, `HEFRoots`
2018 - 2020 28 months	**Assistant Professor** (without tenure), CIMS, NYU, New York, USA *Research in symbolic computation and root-finding* Collaborators: C. Yap (NYU), V. Pan (CUNY) Publications: [4, 5, 6, 7] Software: `Ccluster` Teaching: 125 hours, NYU
2017-2018 10 months	**Scientific assistant**, Technische Universität Kaiserslautern (TUK), Germany *Implementation of a near optimal root clustering algorithm* Collaborators: C. Yap (NYU), V. Pan (CUNY) Publication: [8] Software: `Ccluster`
2014 - 2016 24 months	**Postdoctoral Researcher**, Inria Nancy - Grand Est, France, GAMBLE group *Certified numerical algorithms for computing topology of curves and surfaces* Advisors: G. Moroz and M. Pouget (Inria) Publications: [9, 10, 11, 12, 13] Software: `subdivision_solver`

[4] MAChine leARning for Optimized Numerical methods
[5] Research Institute for Advanced Mathematics
[6] Model and Control
[7] Geometric Algorithms & Models Beyond the Linear & Euclidean realm
[8] City University of New York
[9] Waterloo Maple Incorporation
[10] European Science Foundation, `https://www.esf.org`

2013 - 2014 11 months	**Research and teaching fellow**, UdS, ICube[11] laboratory, France *Reparameterization and homotopy for point distance constraint problems* Advisors: P. Mathis and P. Schreck, (UdS) Publication: [14] Teaching: 180 hours, UdS
2010 - 2013 36 months	**PhD candidate**, UdS, ICube lab., France *Solving geometric constraints by leading an homotopy method by geometry* Advisor: P. Mathis; Supervisor: P. Schreck (UdS) Publications: [15, 16, 17, 18] PhD thesis: [19] Teaching: 3x64 hours, UdS

Software development

2024 -	**SCimBa**[12]: `Python` library, for Physics Informed Machine Learning (PIML). Solves PDE based problems with PINNs, DeepRitz, DeepFlows, NeuralSemiLagrangian, and hybrid (classical - ML based) approaches. Efficient classical (SS-BFGS, SS-Broyden) and physics informed optimizer (natural gradients). Two backends: `Jax` and `Pytorch`. $\simeq$ 20 000 lines of code; lead developper.
2024	**AnaTrot**: `Python` code based on `Pytorch` for radiative transfer direct and inverse problem.
2021 - 2023	**PWPoly**[13]: `C` library implementing algorithms for root finding and fast evaluation of univariate polynomials ([1]), with `Python` interface. Available in `Maple` 2024 through `fsolve` / `RootFinding:-Isolate`[14], `PW` method $\simeq$ 20 000 lines of code; lead developper. **HEFRoots** is the first version of `PWPoly`.
2018 - 2020	**Ccluster**[15]: `C` library for root clustering. Cited in [2, 3, 4, 5, 6, 7, 8]. $\simeq$ 35 000 lines of code; lead developper. Its interface for `Julia`, `Ccluster.jl`, is deprecated.
2014 - 2016	**subdivision_solver**[16]: `SageMath` package for polynomial system solving. Designed for systems of huge polynomials (in *bitsize* and *degree*). Written in `C++`, `Cython`. Described in [12]. $\simeq$ 8 000 lines of code

[11] Engineering science, computer science and imaging laboratory
[12] `https://www.scimba.org`
[13] `https://gitlab.inria.fr/gamble/pwpoly`
[14] `https://www.maplesoft.com/support/help/maple/view.aspx?path=RootFinding%2fIsolate`
[15] `https://github.com/rimbach/Ccluster`
[16] `https://gitlab.inria.fr/gamble/subdiv-solver`

Recent communications

October 2026	Workshop *Fabrique Ton PINNs II*[17], Forum Teratec, Paris. Organized by Groupe Calcul, CNRS. tutorials on PINNs and `SCimBa`[18]
September 2026	Hackathon *AISSAI-JAXHACK : JAX for HPC/AI hybridization*, Paris-Saclay[19]: course *Introduction to JAX for Scientific Computing*
June 2026	Workshop on *Machine Learning and Automatic Differentiation in JAX for Scientific Computing*, Strasbourg[20]: tutorial on PINNs[21] in `Jax`
May 2026	Summer School *Machine Learning + Simulation*, Sophia Antipolis: course and practical session[22] on PIML
December 2025	Workshop *Fabrique Ton PINNs*[23], Sorbonne Université, Paris. Organized by Groupe Calcul, CNRS. tutorials on PINNs and `SCimBa`[24]

Teaching experiences

2024	**Teaching fellow**, UdS, UFR MAI[25], Licence, 35h contact. Practical sessions for *algorithms and coding* in `C++`.
Summer 2018 to Spring 2020	**Responsible and main instructor** of *Mathematical Techniques for Computer Science Applications*, Master's students in Computer Science, CIMS, NYU, 125 hours lectures. Syllabus: Linear algebra, probability, numeric methods for signal processing, computer graphics and machine learning. 1 grader, 1 teaching assistant.
2013 - 2014	**Teaching fellow**, UdS, UFR MAI: 170 hours exercises sessions, 10 hours lectures Responsible of *Algorithm and Coding*, under-graduate students in mathematics, physics and chemistry; 10h lectures in this course unit. Exercises sessions in *graph theory, computers architecture, ...*
2010 - 2014	**Adjunct professor**, UdS, UFR MAI: 3×64 hours exercises sessions in *theory of operating systems*, *computers architecture*, *graph theory*, *algorithms and coding*, ...

[17] `https://groupe-calcul.cnrs.fr/2026-10-atelier-pinns-forum-teratec.html`
[18] `https://rimbach.gitlabpages.inria.fr/fabrique-ton-pinns-2/`
[19] `https://aissai-jaxhack.sciencesconf.org/`
[20] `https://majsc2026.pages.math.unistra.fr/index.html`
[21] `https://rimbach.gitlabpages.inria.fr/ml_ad_jax_sc_pinns_practical/`
[22] `https://project.inria.fr/mlsim2026/program/`
[23] `https://groupe-calcul.cnrs.fr/2025-12-18-atelier-pinns.html`
[24] `https://rimbach.gitlabpages.inria.fr/fabrique-ton-pinns/`
[25] Mathématique et Informatique

Education

October 2013	**PhD** in Computer Science, UdS, France *Solving geometric constraints by leading an homotopy method by geometry* Advisor: P. Mathis; Supervisor: P. Schreck Committee members: D. Michelucci, B. Mourrain, M.Tajine, P. Serré.
June 2010	**Master's degree** in Computer Science, UdS Specialty: Computer Science and Imaging
September 2008	**Master's degree** in Mathematics, UdS Specialty: Discrete Mathematics

Referees

Dr. Emmanuel Franck (`emmanuel.franck@inria.fr`) is researcher at INRIA Nancy-Grand Est, France

Dr. Guillaume Moroz (`guillaume.moroz@inria.fr`) is researcher at INRIA Nancy-Grand Est, France
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Dr. Jürgen Gerhard (`jgerhard@maplesoft.com`) is Senior director of research at Maplesoft, Waterloo Maple Incorporation

Pr. Victor Y. Pan (`victor.pan@lehman.cuny.edu`) is Distinguished professor of Math. at CUNY. Graduate Center of the CUNY, 365 Fifth Avenue, New York, NY 10016-4309, USA

Scientific diffusion

Publications

In the community of symbolic computing where [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 20] have been published, the usage is to make appear the names of the authors in their alphabetical order.

In the community of computer graphics where [14, 15, 16, 17, 18] have been disseminated, the usage is to make appear the name of the main contributor in first position.

International journals

[1] Rémi Imbach, and Guillaume Moroz. Fast evaluation and root finding for polynomials with floating-point coefficients. *Journal of Symbolic Computation*, 127, 2025

[5] Rémi Imbach, Marc Pouget and Chee Yap. Clustering Complex Zeros of Triangular Systems of Polynomials. *Mathematics in Computer Science*, 1–22, 2020 `https://doi.org/10.1007/s11786-020-00482-0`

[9] Rémi Imbach, Guillaume Moroz, and Marc Pouget. Reliable location with respect to the projection of a smooth space curve. *Reliable Computing*, 26:13 – 55, 2018. `https://interval.louisiana.edu/reliable-computing-journal/tables-of-contents.html#Volume_26`

[10] Rémi Imbach, Guillaume Moroz, and Marc Pouget. A certified numerical algorithm for the topology of resultant and discriminant curves. *Journal of Symbolic Computation*, 80, Part 2:285 – 306, 2017 `http://dx.doi.org/10.1016/j.jsc.2016.03.011`

[14] Rémi Imbach, Pascal Mathis, and Pascal Schreck. A robust and efficient method for solving point distance problems by homotopy. *Mathematical Programming*, 163:115–144, 2017 `https://doi.org/10.1007/s10107-016-1058-7`

[15] Rémi Imbach, Pascal Schreck, and Pascal Mathis. Leading a continuation method by geometry for solving geometric constraints. *Computer-Aided Design*, 46:138–147, 2014 `https://doi.org/10.1016/j.cad.2013.08.026`

International conferences, peer reviewed

[2] Rémi Imbach and Victor Y. Pan. Accelerated Subdivision for Clustering Roots of Polynomials Given by Evaluation Oracles. In *Computer Algebra in Scientific Computing: 24th International Workshop, CASC 2022*, 143–164, 2022, Springer. `https://link.springer.com/chapter/10.1007/978-3-031-14788-3_9`

[3] Rémi Imbach and Victor Y. Pan. Root Radii and Subdivision for Polynomial Root-Finding. In *Computer Algebra in Scientific Computing: 23rd International Workshop, CASC 2021*, 136–156, 2021, Springer. `https://link.springer.com/chapter/10.1007/978-3-030-85165-1_9`

[4] Rémi Imbach and Victor Y. Pan. New progress in univariate polynomial root finding. In *International Symposium on Symbolic and Algebraic Computation, ISSAC 2020*, page 249–256, 2020, Association for Computing Machinery. `https://doi.org/10.1145/3373207.3404063`

[6] Rémi Imbach and Victor Y. Pan. New practical advances in polynomial root clustering. In *Mathematical Aspects of Computer and Information Sciences, MACIS 2019*, pages 122–137, Cham, 2019. Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-030-43120-4_11

[7] Rémi Imbach, Victor Y. Pan, Chee Yap, Ilias S. Kotsireas, and Vitaly Zaderman. Root-finding with implicit deflation. In *Computer Algebra in Scientific Computing, CASC 2019*, pages 236–245, Cham, 2019. Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-030-26831-2_16

[8] Rémi Imbach, Victor Y. Pan, and Chee Yap. Implementation of a near-optimal complex root clustering algorithm. In *Mathematical Software, ICMS 2018*, pages 235–244, Cham, 2018. Springer International Publishing. https://link.springer.com/chapter/10.1007/978-3-319-96418-8_28

[11] Rémi Imbach, Guillaume Moroz, and Marc Pouget. Numeric and Certified Isolation of the Singularities of the Projection of a Smooth Space Curve. In *Mathematical Aspects of Computer and Information Sciences: 6th International Conference, MACIS 2015*, pages 78–92. Springer International Publishing, Cham, 2016. http://dx.doi.org/10.1007/978-3-319-32859-1_6

[16] Pascal Mathis, Pascal Schreck, and Rémi Imbach. Decomposition of geometrical constraint systems with reparameterization. In *Proceedings of the 27th Annual ACM Symposium on Applied Computing, SAC 2012*, pages 102–108. ACM, 2012. http://dl.acm.org/citation.cfm?id=2245298

[18] Rémi Imbach, Pascal Mathis, and Pascal Schreck. Tracking method for reparametrized geometrical constraint systems. In *13th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNASC 2011*, pages 31–38. IEEE, 2011. https://ieeexplore.ieee.org/abstract/document/6169498

Research report

[20] Rémi Imbach and Victor Y. Pan. Polynomial root clustering and explicit deflation. arXiv preprint arXiv:1906.04920, June 2019. https://arxiv.org/abs/1906.04920

Technical report

[12] Rémi Imbach. A Subdivision Solver for Systems of Large Dense Polynomials. Technical Report RT-0476, INRIA Nancy, March 2016. https://arxiv.org/abs/1603.07916

National conferences, non peer reviewed

[13] Rémi Imbach, Guillaume Moroz, and Marc Pouget. A certified numerical approach to describe the topology of projected curves. In *Journées de l'Association Française d'Informatique Graphique*, 2015.

[17] Rémi Imbach, Pascal Mathis, and Pascal Schreck. Une approche par décomposition et reparamétrisation de systèmes de contraintes géométriques. In *Journées du Groupe de Travail en Modélisation Géométrique*, 2012.

PhD Thesis

[19] Rémi Imbach. *Résolution de contraintes géométriques en guidant une méthode homotopique par la géométrie*. PhD thesis, Université de Strasbourg, 2013. https://tel.archives-ouvertes.fr/tel-01124176/

Selected communications

Seminars

Jan. 2021	*Complex Roots Clustering*, Séminar PolSys-SpecFun[26], Paris
Nov. 2019	*Practical Advances in Complex Root Clustering.* Joint CUNY Graduate Center-Courant Seminar in Symbolic-Numeric Computing, Courant Institute of Mathematical Sciences, New York. [27]
Jan. 2019	*Complex Roots/Solutions Clustering Algorithms.* OGRE team seminar, Nantes, France
May 2018	*Numerical and certified computation of the topology of projected curves.* Joint CUNY Graduate Center-Courant Seminar in Symbolic-Numeric Computing, CUNY Graduate Center, New York.
July 2017	*Certified numerical tools for computing the topology of projected curves.* AGAG seminar, Technische Universität Kaiserslautern Germany [28]
Sept. 2016	*Certified numerical tools for computing the topology of projected curves.* AriC seminar, ENS(École Normale Supérieure) Lyon, France [29]

[26] https://polsys.lip6.fr/~safey/seminar.html
[27] http://qcpages.qc.cuny.edu/~aovchinnikov/seminar.html
[28] http://www.mathematik.uni-kl.de/en/agag/talks/20170101/
[29] http://www.ens-lyon.fr/LIP/AriC/seminar

International conferences and workshops

July 2023	*Fast evaluation and root finding for polynomials with floating-point coefficients*. ISSAC 2023 (International Symposium on Symbolic and Algebraic Computation), Tromso, Norway conference[30].
July 2020	*New progress in univariate polynomial root finding*. ISSAC 2020, online conference[31].
Aug. 2019	*Clustering Complex Zeros of Triangular Systems of Polynomials*. CASC 2019 (Computer Algebra in Scientific Computing), Moscow, Russia[32].
July 2018	*Implementation of a near-optimal complex root clustering algorithm*. ICMS 2018 (International Congress of Mathematical Software), Notre-Dame, USA[33]
June 2016	*Interval tools for computing the topology of projected curves*. SWIM 2016 (Summer Workshop on Interval Methods), Lyon, France[34]
Novembre 2015	*Numeric and Certified Isolation of the Singularities of the Projection of a Smooth Space Curve*. MACIS 2015 (Sixth International Conference on Mathematical Aspects of Computer and Information Science), Berlin, Germany[35]
Novembre 2013	*Leading a continuation method by geometry for solving geometric constraints*. GD/SPM 13 (Geometric and Physical Modeling), Denver, Colorado, USA
Septembre 2011	*Tracking method for reparametrized geometrical constraint systems*. SYNASC 11 (Symposium on Symbolic and Numeric Algorithms for Scientific Computing), Timisoara, Romania

[30] http://www.issac-conference.org/2023/
[31] http://www.issac-conference.org/2020/
[32] http://www.casc-conference.org/2019/
[33] http://icms-conference.org/2018/
[34] https://swim2016.sciencesconf.org/
[35] http://macis2015.zib.de/